

LAB PRACTICE 1

OBJECTIVE

Basic concepts and creation of virtual machines for machine learning

PREREQUISITES

- Lab teams consist of **two students**. Each student uses its own Google account. The accounts are referred to as **User-A and User-B** throughout this document
- Students navigate to **console.cloud.google.com** and login with their account credentials
- Students must have the **Google CLI locally installed**

Note: unless otherwise stated, please use the Google Cloud Console to implement the tasks and leave the configuration defaults

Task 1: create a Compute Engine instance tailored for machine learning

1. Create a **project called "lab1-a"**. The first step to create a Compute Engine instance is to create a virtual private network (VPC)

Are there any already created VPCs? Which one? In what regions is there a subnet?

- create a network called **ml-network** in custom creation mode
- create a subnet in region **europe-west1**
- configure the following subnet range: 10.0.0.0/24

What are the default firewall rules? What are the proposed firewall rules? Do you see any issue with the proposed firewall rules?

Do not select any of the proposed firewall rules

2. Create a new firewall rule called "ml-network-allow-ssh-from-iap" to allow SSH access only from IAP servers:

- targets: All instances in the network
- source IPv4 ranges: 35.235.240.0/20
- ports: TCP 22

3. Create a Compute Engine instance connected to ml-network

- name: **vm1**
- machine type: **n1-standard-1**
- region: **europe-west1**
- zone: **c**
- public image: select **Deep Learning VM M129**
- network tag: **deep**
- network: **ml-network**

Note: list of available Deep Learning images is at <https://cloud.google.com/deep-learning-vm/docs/images>

Connect to the instance using the web SSH option in the console. What are the installed versions of:

- Operating System
- Python
- numpy, scipy and scikit-learn

4. Bring up a cloud shell. Create a SSH tunnel towards the Compute Engine instance.

```
gcloud compute ssh --project PROJECT_ID --zone ZONE vm1 -- -L 8080:localhost:8080
```

NOTE: make sure that firewall rules are updated to allow this flow

5. Connect to Jupyter notebook through the Web Preview on port 8080.

What is the installed version of JupyterLab?

Task 2: use the Cloud Shell. Create a cloud storage bucket

Note: for this task we will use the Google Cloud CLI from the Cloud Shell machine

6. Check what are the CloudShell properties defined in the current configuration

```
gcloud config list
```

7. Set default values for the PROJECT if not already set

```
gcloud config set project PROJECT
```

8. Create a Cloud Storage bucket

```
gcloud storage buckets create gs://[BUCKET_NAME] --location europe-west1
```

9. Verify in the Cloud Console that it has been created

What is the bucket location type? What does it mean?

What is the bucket storage class? What does it mean?

10. Upload a file from your local PC to the CloudShell machine using the Cloud Shell toolbar

11. Copy the file to the created Cloud Storage bucket

```
gcloud storage cp [FILE] gs://[BUCKET_NAME]
```

12. Access your file from the browser in your PC

What is needed to access the file?

Task 3: creating and using custom images

13. Create a Compute Engine instance connected to ml-network

- name: `vm2`
- machine type: `e2-medium`
- region: `eu-west-1`
- public image: leave the default `Debian 12`
- network tag: `deep`
- network: `ml-network`

14. Connect to the instance using the web SSH option in the console. Install the nginx software:

```
sudo apt update
sudo apt install nginx
```

15. Verify the installation by pointing a browser to vm2

16. Create an image of vm2's boot disk

- name: `nginx`

17. Delete Compute Engine instance vm2 and associated boot disk

18. Create a new Compute Engine instance `vm3`

- name: `vm3`
- machine type: `e2-medium`
- region: `eu-west-1`
- custom image: `my-nginx`
- network tag: `deep`
- network: `ml-network`

19. Verify the installation by pointing a browser to vm3

20. Delete all the Compute Engine, Disks and Image resources created during the lab practice. Do not delete the network since it will be reused in subsequent lab practices

Task 4: explore Google Cloud Market place

21. Explore Cloud Marketplace

Find some AI and machine learning solutions in the Marketplace and provide their names Which are the advantages of using the Marketplace solutions?

Task 5: [OPTIONAL] Implement Task 3 using the Google CLI

> Provide the list of gcloud commands used

Task 6: [OPTIONAL] Implement Task 3 using the Google API

> Provide a program that uses the Google API and implements Task 3. Use the available Google provided SDKs for the most popular languages